AP CALCULUS AB – UNIT 3

3.3 Inverse Function and Inverse Trigonometric Derivatives

Name: ____________ **Class:** ____________ **Date:** ____________

Lesson Overview and Learning Objectives

- Use the derivative-of-an-inverse formula from equations, tables, and graphs.
- Differentiate inverse trigonometric functions and compositions containing them.
- Write tangent lines to inverse functions and interpret reciprocal slopes.
- Combine inverse derivatives with the Chain, Product, and Quotient Rules.

Keys: Essential Knowledge and Vocabulary

Inverse derivative	If $f(a) = b$ and $f'(a) \neq 0$, then $(f^{-1})'(b) = 1/f'(a) = 1/f'(f^{-1}(b))$.
Reciprocal slope	A function and its inverse have reciprocal tangent slopes at reflected points.
Arcsine	$\frac{d}{dx}\arcsin u = \frac{u'}{\sqrt{1-u^2}}$.
Arccosine	$\frac{d}{dx}\arccos u = -\frac{u'}{\sqrt{1-u^2}}$.
Arctangent	$\frac{d}{dx}\arctan u = \frac{u'}{1+u^2}$.
Domain awareness	Square-root denominators require valid inputs and may vanish at domain endpoints.

Explanation, Strategy, and Common Errors

- To find $(f^{-1})'(b)$, first identify the preimage a satisfying $f(a) = b$.
- Evaluate f' at the preimage, not at the output b, then take the reciprocal.
- For inverse trigonometric compositions, multiply the standard derivative by the derivative of the inside.
- In a table, locate the output in the $f(x)$ row before reading the corresponding derivative entry.
- Use radians for trigonometric derivative formulas.

Solved Example: Inverse derivative from a formula

Let $f(x) = x^3 + x$. Find $(f^{-1})'(2)$.

Answer and Reasoning

Since $f(1) = 2$ and $f'(1) = 4$, $(f^{-1})'(2) = 1/4$.

Solved Example: Arcsine composition

Differentiate $y = \arcsin(3x)$.

Answer and Reasoning

$y' = \dfrac{3}{\sqrt{1-9x^2}}$.

Solved Example: Arctangent composition

Differentiate $y = \arctan(x^2)$.

Answer and Reasoning

$y' = \dfrac{2x}{1+x^4}$.

Shared Representation / Data

x	-2	-1	0	1	3
$f(x)$	-5	-2	1	4	10
$f'(x)$	4	3	2	5	7

AP Exam Language

For every derivative in context, include the sign, units, input value, and a complete verbal interpretation. For every existence claim, state the relevant hypothesis or one-sided comparison.

Leveled Practice

Easy Foundations

Build fluency with the definition, notation, and one-step applications.

1. If $f(2) = 7$ and $f'(2) = 5$, find $(f^{-1})'(7)$.

2. Differentiate $y = \arcsin x$.

3. Differentiate $y = \arccos(2x)$.

4. Differentiate $y = \arctan(5x)$.

Medium Applications

Connect representations and apply the lesson procedure in familiar settings.

5. Let $f(x) = x^3 + 2x$. Find $(f^{-1})'(3)$.

6. Write the tangent line to $y = \arcsin x$ at $x = 1/2$.

7. Differentiate $y = \arctan(\sqrt{x})$.

8. Using the table, find $(f^{-1})'(4)$.

Hard Reasoning

Select a method, justify conclusions, and combine several ideas.

9. Differentiate $y = \arcsin\left(\frac{x}{1+x}\right)$.

10. Differentiate $y = x \arctan x$.

11. Find the second derivative of $\arctan x$ at $x = 1$.

12. If $g = f^{-1}$, $f(3) = 8$, and $f'(3) = -2$, find $g'(8)$.

Difficult Analysis

Work through less-direct algebra, subtle hypotheses, and multi-step reasoning.

13. Find a so the tangent to $y = \arctan(ax)$ at $x = 0$ has slope 6.

14. Let $h(x) = f^{-1}(x^2)$. Express $h'(x)$.

15. Explain why $f'(a) = 0$ prevents the usual inverse-derivative formula from giving a finite value at $f(a)$.

Challenge / AP Extension

Synthesize the topic in unfamiliar, proof-oriented, or parameter-based problems.

16. Prove the inverse-derivative formula from $f(f^{-1}(x)) = x$.

17. Differentiate and simplify $y = \arctan\left(\frac{1-x}{1+x}\right)$.

AP-Style Multiple-Choice Practice

Part A: No Calculator

Choose the best answer. Justification is not required on the AP multiple-choice section, but show reasoning while practicing.

1. If $f(4) = 9$ and $f'(4) = -3$, then $(f^{-1})'(9)$ is

(A) -3
(B) $-1/3$
(C) $1/3$
(D) 3

2. The derivative of $\arctan(2x)$ is

(A) $2/(1+4x^2)$
(B) $1/(1+2x^2)$
(C) $2/\sqrt{1-4x^2}$
(D) $-2/(1+4x^2)$

3. Using the table, $(f^{-1})'(10)$ equals

(A) $1/7$
(B) $1/10$
(C) 7
(D) 10

4. A function and its inverse have tangent slopes that are

(A) equal
(B) negative reciprocals
(C) reciprocals at reflected points
(D) always positive

Part B: Graphing Calculator Permitted

Use a calculator only for numerical evaluation, graphing, or regression as indicated. Preserve mathematical setup.

5. Let $f(x) = x + e^x$. If $f(a) = 4$, then $(f^{-1})'(4)$ is closest to

(A) 0.119
(B) 0.255
(C) 0.500
(D) 0.731

6. At $x = 0.6$, the derivative of $y = \arcsin(x^2)$ is closest to

(A) 0.75
(B) 1.29
(C) 1.67
(D) 2.08

AP-Style Free-Response Practice

Calculator Permitted

Let $f(x) = x + e^x$, which is strictly increasing, and let $g = f^{-1}$.

(a) Solve $f(a) = 4$ numerically. [2 points]

(b) Find $g'(4)$ to three decimals. [2 points]

(c) Write the tangent line to g at $(4, a)$. [2 points]

No Calculator

The table gives values of a differentiable one-to-one function f. Let $g = f^{-1}$.

(a) Find $g(4)$ and $g'(4)$. [2 points]

(b) Find the derivative of $H(x) = \arctan(g(x))$ at $x = 4$. [3 points]

(c) Explain geometrically why $g'(4)$ is positive. [2 points]